

Synchronization, Phase Locking, and Amplitude Death in Coupled Stuart-Landau Oscillators

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The Stuart-Landau oscillator is a canonical nonlinear model for studying self-sustained oscillations near a supercritical Hopf bifurcation. In this work, we analyze the model in three stages. First, the single Stuart-Landau oscillator is studied in Cartesian and polar coordinates. The Cartesian form gives the real planar system and the Jacobian at the origin, while the polar form separates amplitude and phase and shows that a stable limit cycle of radius $r^* = \sqrt{\lambda}$ exists for $\lambda > 0$. Second, we study two diffusively coupled Stuart-Landau oscillators with different natural frequencies. A phase reduction gives the Adler-type equation $\dot{\phi} = \Delta\omega - 2K \sin \phi$, which predicts the phase-locking threshold $K_c = |\Delta\omega|/2$ when amplitudes remain close to the limit cycle. The full coupled system is also used to study amplitude death, where coupling and frequency mismatch can stabilize the origin and suppress oscillations. Third, the model is extended to networks with chain, ring, global, feedforward, and general graph topologies. Network synchronization is measured using an instantaneous Kuramoto-style order parameter $R(t)$, a time-averaged order parameter $\bar{R}$, and a frequency-spread diagnostic. The original contribution of this project is a numerical comparison of phase drift, phase locking, amplitude death, and topology-dependent synchronization using a common set of diagnostics. The simulations show that global coupling synchronizes most efficiently, while chain coupling requires the largest coupling strength.

I. INTRODUCTION

Oscillatory behavior is one of the most common patterns in applied mathematics, physics, biology, chemistry, and engineering. A system is called an oscillator when it repeatedly returns near previous states as time evolves. The motion of a pendulum, the voltage in an electrical circuit, the firing of neurons, the rhythm of a heartbeat, and the electromagnetic field in a laser are all examples of oscillatory systems. In many applications, however, the most important behavior is not just that one system oscillates. The more interesting question is what happens when many oscillators interact.

When oscillators are coupled, they can display collective behavior that would not appear in a single isolated oscillator. Two oscillators with different natural frequencies may eventually rotate at the same average frequency if the coupling between them is strong enough. A large network of oscillators may start with disordered phases and later organize into a coherent rhythm. In some cases, coupling can even eliminate oscillations entirely. These outcomes are examples of synchronization, phase locking, and amplitude death.

The Stuart-Landau oscillator is a natural model for studying these ideas. It is simple enough that much of the analysis can be done by hand, but it is rich enough to exhibit nonlinear behavior. The single Stuart-Landau oscillator is written in complex form as

$$\dot{z} = (\lambda + i\omega - |z|^2)z, \quad (1)$$

where $z(t) \in \mathbb{C}$ is the oscillator state, $\lambda \in \mathbb{R}$ is the bifurcation parameter, and $\omega \in \mathbb{R}$ is the natural angular frequency. The parameter λ controls growth or decay near the origin, while ω controls rotation. The nonlinear

term $-|z|^2z$ saturates growth and creates a stable finite-amplitude oscillation.

One reason Eq. (1) is important is that it is the normal form for a supercritical Hopf bifurcation. Near a Hopf bifurcation, many complicated nonlinear systems can be reduced to an equation with the same amplitude structure as the Stuart-Landau oscillator [1, 4]. Thus, even though the model is simple, it captures a universal mechanism for the birth of stable oscillations.

This project follows three stages. First, we analyze the single oscillator and show that it has a stable limit cycle when $\lambda > 0$. Second, we introduce two coupled Stuart-Landau oscillators and derive a critical coupling condition for phase locking. Third, we extend the model to networks of many oscillators and compare how different coupling topologies affect synchronization. The analytical reductions used in the single- and two-oscillator cases are standard results from nonlinear dynamics and synchronization theory [2, 3]. The original component of this project is the numerical comparison of phase drift, phase locking, amplitude death, and topology-dependent synchronization using a common set of diagnostics: amplitudes, phase differences, instantaneous order parameter, time-averaged order parameter, and frequency spread.

II. SINGLE STUART-LANDAU OSCILLATOR

The single Stuart-Landau oscillator is

$$\dot{z} = (\lambda + i\omega - |z|^2)z. \quad (2)$$

Although Eq. (2) is written as one complex-valued equation, it is equivalent to a two-dimensional real dynamical

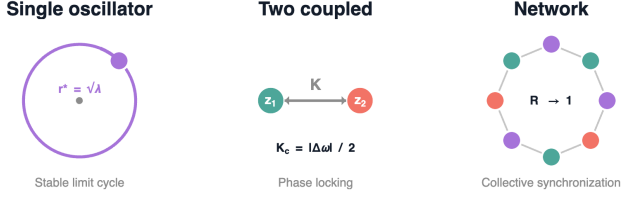


FIG. 1. Project overview. The study begins with the single Stuart-Landau oscillator, then moves to two coupled oscillators, and finally extends to networks of many interacting oscillators. This progression is useful because each stage adds one new layer of behavior: limit cycles, phase locking, and network synchronization.

system. Write

$$z = x + iy,$$

where $x(t)$ and $y(t)$ are real-valued functions of time. In this interpretation, x and y are coordinates in the phase plane. The amplitude of the oscillator is

$$|z| = \sqrt{x^2 + y^2},$$

and the phase is the angle of the point (x, y) relative to the positive x -axis.

The equation contains three important pieces. The term λz is responsible for linear growth or decay. If $\lambda < 0$, small perturbations shrink and the origin is attracting. If $\lambda > 0$, small perturbations grow and the origin is repelling. The term $i\omega z$ produces rotation. The term $-|z|^2 z$ is nonlinear damping. It is weak near the origin, but it becomes stronger as the amplitude increases. This nonlinear damping prevents unbounded growth and produces a stable amplitude.

A. Cartesian Form

To rewrite Eq. (2) in real variables, substitute $z = x + iy$. Since

$$|z|^2 = x^2 + y^2$$

and

$$\dot{z} = \dot{x} + i\dot{y},$$

we obtain

$$\dot{x} + i\dot{y} = (\lambda + i\omega - x^2 - y^2)(x + iy).$$

Expanding and collecting real and imaginary parts gives

$$\dot{x} = (\lambda - x^2 - y^2)x - \omega y, \quad (3)$$

$$\dot{y} = (\lambda - x^2 - y^2)y + \omega x. \quad (4)$$

This form is useful because it allows the standard tools of planar dynamical systems to be applied. The terms involving ω rotate the vector field, while the terms involving $\lambda - x^2 - y^2$ control radial growth or decay.

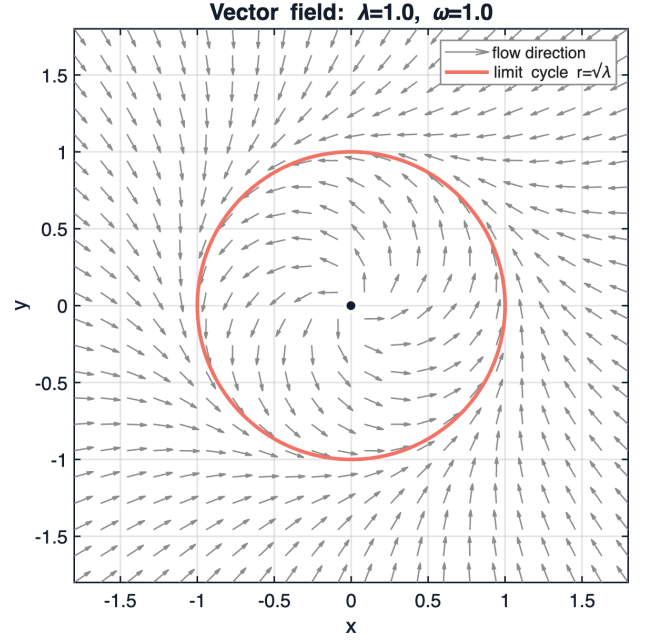


FIG. 2. Cartesian phase-plane representation of the single Stuart-Landau oscillator. The vector field combines rotation from the ω terms with radial growth or decay from the $\lambda - x^2 - y^2$ terms.

B. Equilibria and Jacobian Linearization

An equilibrium satisfies

$$\dot{x} = 0, \quad \dot{y} = 0.$$

The origin $(x, y) = (0, 0)$ is clearly an equilibrium. For the usual case $\omega \neq 0$, it is the only equilibrium. To see this, write Eqs. (3)–(4) in matrix form:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \lambda - r^2 & -\omega \\ \omega & \lambda - r^2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad r^2 = x^2 + y^2. \quad (5)$$

A nonzero equilibrium would require the determinant of the matrix in Eq. (5) to vanish:

$$(\lambda - r^2)^2 + \omega^2 = 0.$$

For $\omega \neq 0$, this cannot occur.

The Jacobian matrix for the real planar system is

$$J(x, y) = \begin{pmatrix} \lambda - 3x^2 - y^2 & -2xy - \omega \\ -2xy + \omega & \lambda - x^2 - 3y^2 \end{pmatrix}.$$

At the origin,

$$J(0, 0) = \begin{pmatrix} \lambda & -\omega \\ \omega & \lambda \end{pmatrix}.$$

The eigenvalues are

$$\eta_{\pm} = \lambda \pm i\omega.$$

The real part of both eigenvalues is λ . If $\lambda < 0$, the origin is a spiral sink. If $\lambda > 0$, the origin is a spiral source. If $\lambda = 0$, the linearization has purely imaginary eigenvalues. This eigenvalue calculation is the first sign of a Hopf bifurcation.

C. Polar Form and Amplitude Dynamics

The polar form gives the clearest explanation of the nonlinear dynamics. Let

$$z = re^{i\theta},$$

where $r(t) \geq 0$ is the amplitude and $\theta(t)$ is the phase. Then

$$\dot{z} = (\dot{r} + ir\dot{\theta})e^{i\theta}.$$

Substituting into Eq. (2) gives

$$(\dot{r} + ir\dot{\theta})e^{i\theta} = (\lambda + i\omega - r^2)re^{i\theta}.$$

Canceling $e^{i\theta}$ and equating real and imaginary parts yields

$$\begin{aligned}\dot{r} &= (\lambda - r^2)r, \\ \dot{\theta} &= \omega.\end{aligned}$$

The radial dynamics and angular dynamics separate completely. The amplitude equation determines whether the trajectory moves toward or away from the origin, while the phase equation says that the oscillator rotates with constant angular frequency ω .

The amplitude equation is

$$\dot{r} = r(\lambda - r^2).$$

The radial equilibria satisfy

$$r = 0 \quad \text{or} \quad r = \sqrt{\lambda}.$$

The second equilibrium exists only for $\lambda \geq 0$. If

$$F(r) = \lambda r - r^3,$$

then

$$F'(r) = \lambda - 3r^2.$$

Thus

$$F'(0) = \lambda, \quad F'(\sqrt{\lambda}) = -2\lambda.$$

Therefore $r = 0$ is stable for $\lambda < 0$ and unstable for $\lambda > 0$, while $r = \sqrt{\lambda}$ is stable for $\lambda > 0$.

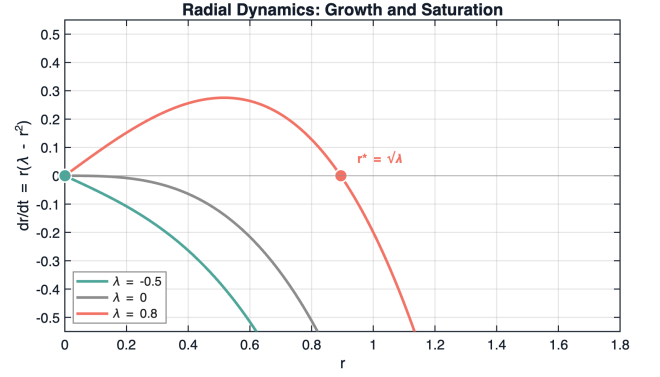


FIG. 3. Radial dynamics of the single oscillator. For $\lambda > 0$, the stable radial equilibrium $r = \sqrt{\lambda}$ corresponds to the stable limit cycle in the phase plane.

D. Phase Dynamics and Limit Cycle

The phase equation is

$$\dot{\theta} = \omega.$$

Integrating gives

$$\theta(t) = \theta(0) + \omega t.$$

Thus the oscillator rotates uniformly. The parameter ω controls the angular speed, and its sign controls the direction of rotation.

When $\lambda > 0$, the long-time solution has the approximate form

$$z(t) \approx \sqrt{\lambda} e^{i(\omega t + \theta_0)}.$$

In Cartesian coordinates, the stable radius corresponds to the circle

$$x^2 + y^2 = \lambda, \quad \lambda > 0.$$

Since trajectories near the circle approach it and then rotate around it, this circle is a stable limit cycle.

E. Hopf Bifurcation

A Hopf bifurcation occurs when a pair of complex conjugate eigenvalues crosses the imaginary axis and a periodic orbit is created or destroyed. For the Stuart-Landau oscillator, the eigenvalues at the origin are

$$\eta_{\pm} = \lambda \pm i\omega,$$

so

$$\text{Re}(\eta_{\pm}) = \lambda.$$

As λ passes through zero, the real part changes sign. For $\lambda < 0$, trajectories spiral into the origin. For $\lambda > 0$, trajectories spiral away from the origin.

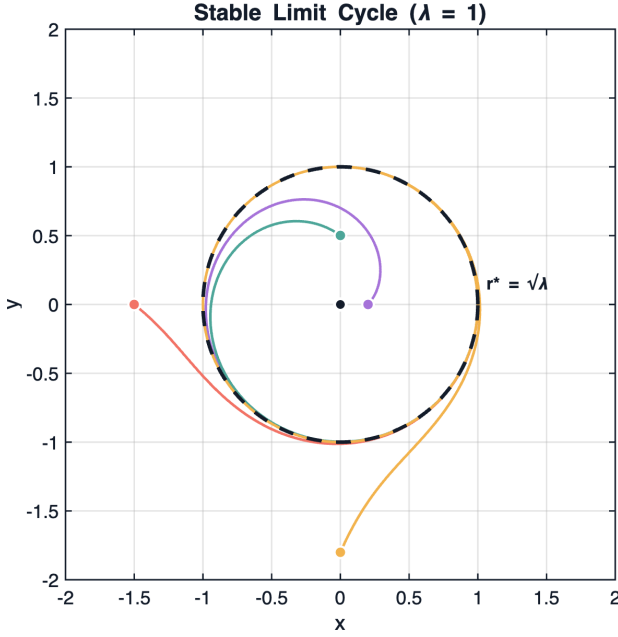


FIG. 4. Stable limit cycle for the single Stuart-Landau oscillator. For $\lambda > 0$, trajectories approach the circle $r = \sqrt{\lambda}$ while rotating with angular frequency ω .

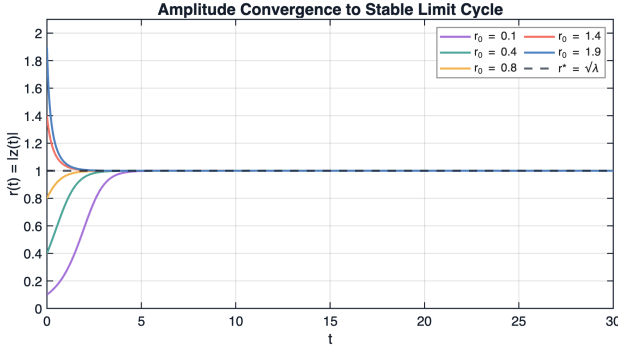


FIG. 5. Amplitude convergence for the single oscillator. Initial conditions inside and outside the limit cycle both approach the same stable radius $r = \sqrt{\lambda}$.

The nonlinear radial equation tells us what replaces the stable origin after it loses stability. For $\lambda > 0$, the stable amplitude is

$$r^* = \sqrt{\lambda}.$$

Therefore, a stable periodic orbit appears for positive λ . Because the stable limit cycle exists for $\lambda > 0$, the bifurcation is supercritical. The cubic term is what makes the limit cycle stable. For small r and $\lambda > 0$, the term λr dominates, so the radius grows. For large r , the term $-r^3$ dominates, so the radius shrinks. The balance between these two effects produces the stable radius $\sqrt{\lambda}$.

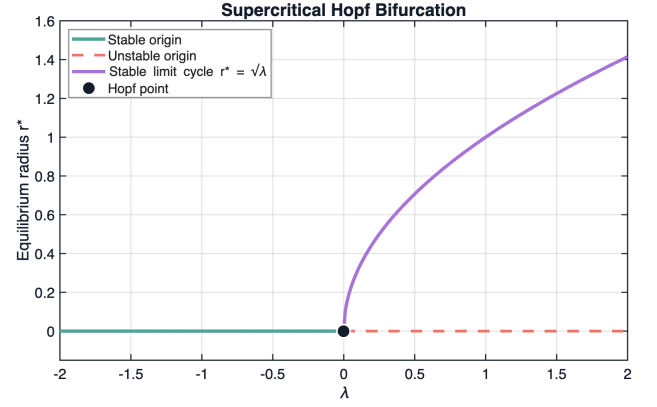


FIG. 6. Supercritical Hopf bifurcation diagram. For $\lambda < 0$, the origin is stable. For $\lambda > 0$, the origin becomes unstable and a stable limit cycle of radius $r = \sqrt{\lambda}$ appears.

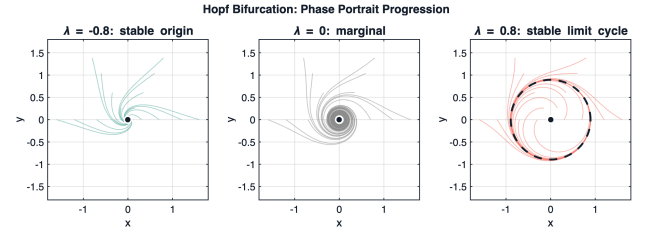


FIG. 7. Phase portrait view of the Hopf bifurcation. The origin changes stability as λ crosses zero, and a stable periodic orbit appears for positive λ .

III. TWO COUPLED STUART-LANDAU OSCILLATORS

The second stage is to introduce coupling between two oscillators. This is the simplest setting in which synchronization, phase locking, and amplitude death can be studied.

A. Coupled Model

We consider two diffusively coupled Stuart-Landau oscillators:

$$\dot{z}_1 = (\lambda + i\omega_1 - |z_1|^2) z_1 + K(z_2 - z_1), \quad (6)$$

$$\dot{z}_2 = (\lambda + i\omega_2 - |z_2|^2) z_2 + K(z_1 - z_2). \quad (7)$$

Here $K \geq 0$ is the coupling strength. The terms $K(z_2 - z_1)$ and $K(z_1 - z_2)$ pull each oscillator toward the other. This is called diffusive coupling because it reduces differences between oscillator states.

The natural frequency mismatch is

$$\Delta\omega = \omega_2 - \omega_1.$$

If $K = 0$ and $\Delta\omega \neq 0$, the oscillators rotate at different rates. Their phase difference grows over time. Coupling

can potentially overcome this mismatch, but only if K is large enough.

B. Phase Difference

Write

$$z_j = r_j e^{i\theta_j}, \quad j = 1, 2.$$

The phase difference is

$$\phi(t) = \theta_2(t) - \theta_1(t).$$

If $\phi(t)$ approaches a constant, then the oscillators are phase locked:

$$\phi(t) \rightarrow \phi^*.$$

If $\phi(t)$ does not settle, then the oscillators exhibit phase drift. This distinction is important. Two oscillators can both remain on stable limit cycles but still fail to synchronize. Synchronization is about timing and relative phase, not just about the existence of oscillations.

C. Phase Reduction

Starting from Eqs. (6)–(7), write

$$z_j = r_j e^{i\theta_j}.$$

Then

$$\dot{z}_j = (\dot{r}_j + ir_j \dot{\theta}_j) e^{i\theta_j}.$$

For oscillator 1,

$$(\dot{r}_1 + ir_1 \dot{\theta}_1) e^{i\theta_1} = (\lambda + i\omega_1 - r_1^2) r_1 e^{i\theta_1} + K(r_2 e^{i\theta_2} - r_1 e^{i\theta_1}).$$

Multiplying by $e^{-i\theta_1}$ and using $\phi = \theta_2 - \theta_1$ gives

$$\begin{aligned} \dot{r}_1 &= (\lambda - r_1^2) r_1 + K(r_2 \cos \phi - r_1), \\ \dot{\theta}_1 &= \omega_1 + K \frac{r_2}{r_1} \sin \phi. \end{aligned}$$

For oscillator 2, the same calculation gives

$$\begin{aligned} \dot{r}_2 &= (\lambda - r_2^2) r_2 + K(r_1 \cos \phi - r_2), \\ \dot{\theta}_2 &= \omega_2 - K \frac{r_1}{r_2} \sin \phi. \end{aligned}$$

Therefore

$$\dot{\phi} = \omega_2 - \omega_1 - K \left(\frac{r_1}{r_2} + \frac{r_2}{r_1} \right) \sin \phi. \quad (8)$$

This equation is still coupled to the amplitude equations. If the coupling does not strongly disturb the amplitudes, then

$$r_1 \approx r_2 \approx \sqrt{\lambda}.$$

Thus

$$\frac{r_1}{r_2} \approx 1, \quad \frac{r_2}{r_1} \approx 1.$$

Substituting into Eq. (8) gives the reduced phase equation

$$\dot{\phi} = \Delta\omega - 2K \sin \phi, \quad \Delta\omega = \omega_2 - \omega_1. \quad (9)$$

The term $\Delta\omega$ represents the natural tendency of the oscillators to drift apart in phase. The term $-2K \sin \phi$ represents the tendency of coupling to pull the phases into a fixed relationship.

D. Critical Coupling and Phase Locking

A phase-locked state occurs when

$$\dot{\phi} = 0.$$

Using Eq. (9), this gives

$$\Delta\omega - 2K \sin \phi^* = 0,$$

so

$$\sin \phi^* = \frac{\Delta\omega}{2K}. \quad (10)$$

A real solution exists only if

$$\left| \frac{\Delta\omega}{2K} \right| \leq 1.$$

Therefore

$$K_c = \frac{|\Delta\omega|}{2}.$$

This formula says that the coupling strength needed for phase locking grows with frequency mismatch.

To determine stability, let

$$F(\phi) = \Delta\omega - 2K \sin \phi.$$

A locked state ϕ^* is stable if

$$F'(\phi^*) < 0.$$

Since

$$F'(\phi) = -2K \cos \phi,$$

stability requires

$$\cos \phi^* > 0.$$

The branch with $\cos \phi^* < 0$ is unstable. At $K = K_c$, the stable and unstable locked states collide in a saddle-node bifurcation of the reduced phase equation.

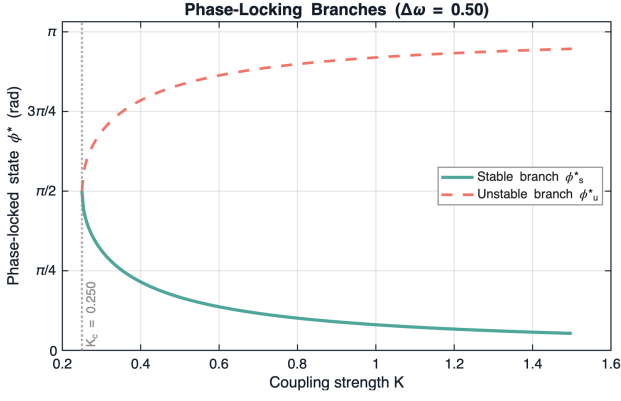


FIG. 8. Stable and unstable phase-locked branches of the reduced phase model for fixed frequency mismatch. The critical coupling $K_c = |\Delta\omega|/2$ marks the saddle-node bifurcation where the stable and unstable locked states first appear. For $K > K_c$, the stable branch gives the attracting phase difference predicted by the reduced phase equation.

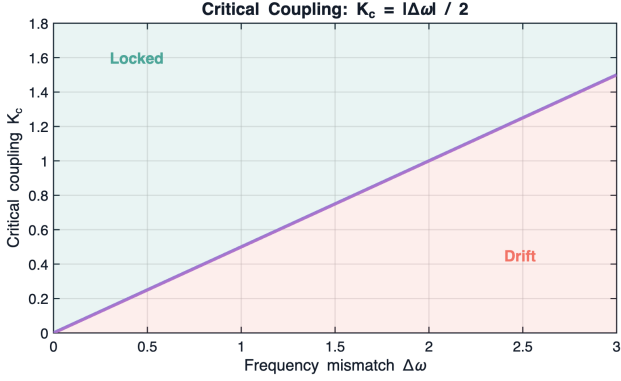


FIG. 9. Critical coupling as a function of frequency mismatch. The reduced phase model predicts the linear relation $K_c = |\Delta\omega|/2$, so larger frequency mismatch requires stronger coupling before phase locking is possible. The shaded regions separate the predicted drifting and locked regimes.

E. Weak and Strong Coupling

The reduced phase model predicts two main behaviors: phase drift for $K < K_c$ and phase locking for $K > K_c$.

For weak coupling,

$$K < K_c.$$

In this case, coupling is not strong enough to balance the natural frequency mismatch. The reduced phase equation has no fixed point because Eq. (10) would require $|\sin \phi^*| > 1$. Numerically, the amplitudes remain close to the limit-cycle radius, but the phase difference does not approach a constant.

For strong coupling,

$$K > K_c.$$

Now the coupling is strong enough to overcome the mis-

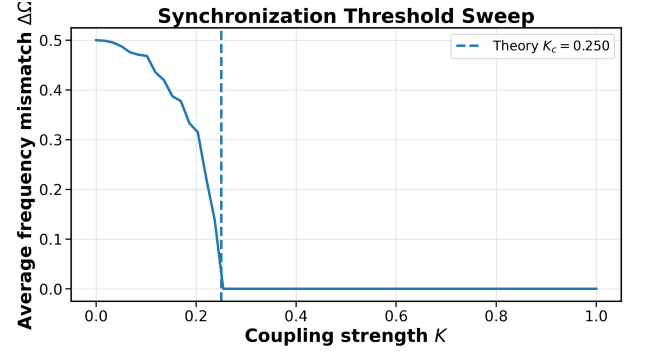


FIG. 10. Numerical sweep of the phase-locking transition as the coupling strength K is varied. The diagnostic shown is the average frequency mismatch after transients have decayed. For small K , the oscillators drift and the measured frequency mismatch remains nonzero. Once K exceeds the locking threshold, the long-time frequency mismatch approaches zero, indicating phase locking. This transition is consistent with the reduced phase prediction $K_c = |\Delta\omega|/2$.

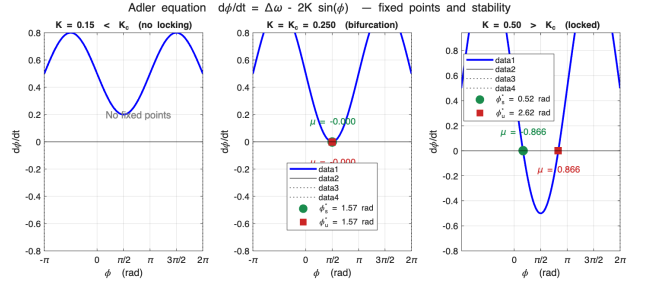


FIG. 11. Stability of the phase-locked states in the reduced phase equation. Small perturbations near the stable locked phase return toward the equilibrium phase difference, while perturbations near the unstable locked phase move away. This behavior agrees with the linear stability condition $F'(\phi^*) = -2K \cos \phi^* < 0$, so the locked branch with $\cos \phi^* > 0$ is stable and the branch with $\cos \phi^* < 0$ is unstable.

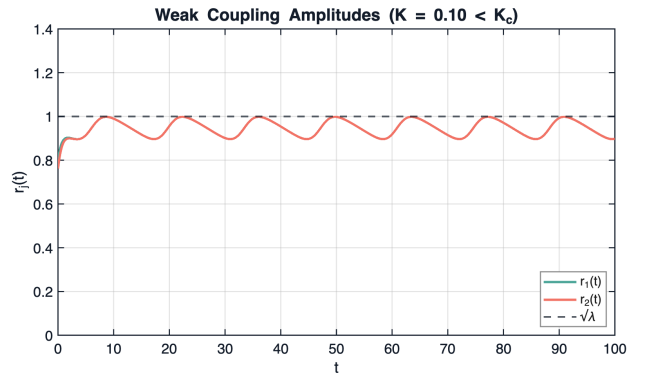


FIG. 12. Weak coupling amplitudes. The amplitudes remain close to the stable limit-cycle radius, showing that both oscillators continue to oscillate even though they are not phase locked.

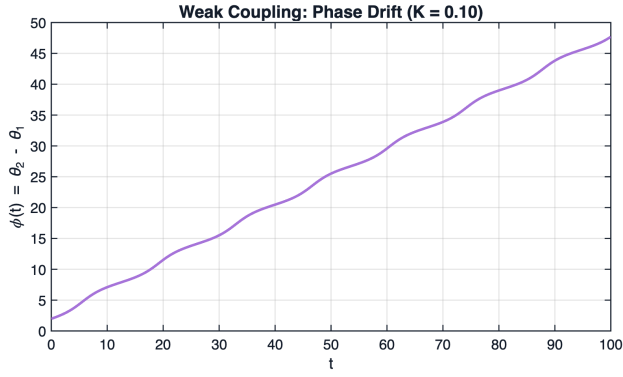


FIG. 13. Weak coupling phase difference. Since $K < K_c$, the coupling is too weak to overcome the frequency mismatch, and the phase difference does not approach a constant. This drifting phase difference is the numerical signature of failure to phase lock.

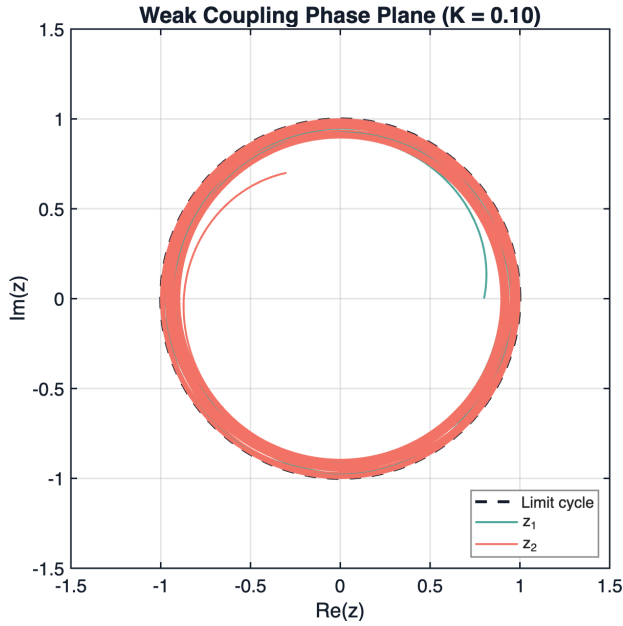


FIG. 14. Weak coupling phase-plane behavior. The oscillators continue to move on stable oscillatory trajectories, but their relative phase does not settle. Thus the amplitudes remain bounded while the timing between oscillators keeps drifting.

match. The phase difference approaches a constant,

$$\phi(t) \rightarrow \phi^*.$$

The oscillators then rotate with the same average frequency. They may not be exactly in phase, but their phase relationship becomes fixed.

F. Amplitude Death

Coupling does not always produce synchronization. In some regimes, coupling can suppress oscillations entirely.

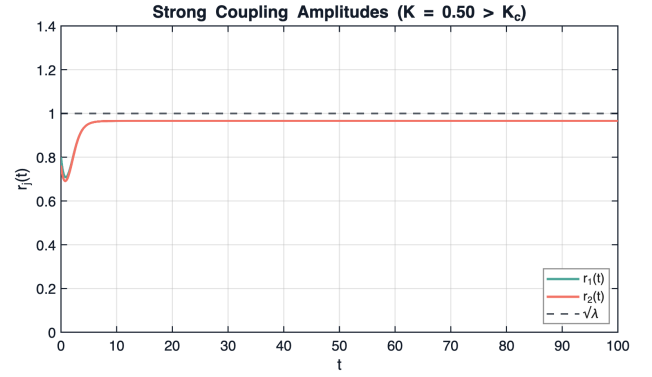


FIG. 15. Strong coupling amplitudes. The amplitudes remain near the stable limit-cycle radius while the phases organize into a locked relationship. This shows that phase locking does not require the oscillators to collapse to the origin.

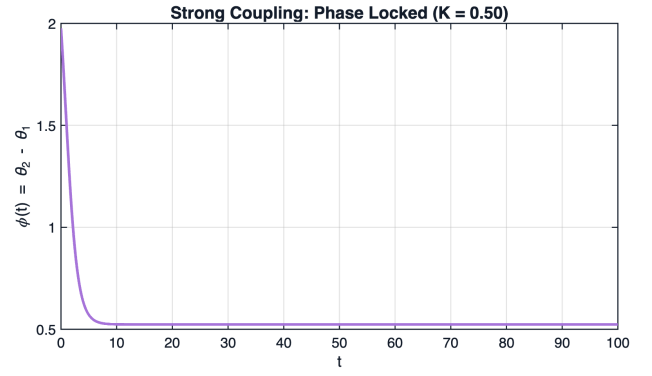


FIG. 16. Strong coupling phase difference. The phase difference approaches a constant, showing that the oscillators have become phase locked. In this regime, both oscillators rotate with the same long-time average frequency.

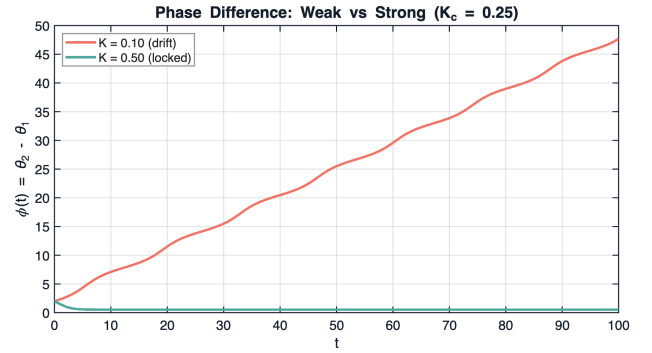


FIG. 17. Comparison of weak and strong coupling. Weak coupling produces persistent phase drift, while strong coupling produces convergence of the phase difference to a constant. This contrast illustrates the transition predicted by the reduced phase model.

This phenomenon is called amplitude death [5]. Instead of approaching a limit cycle, both amplitudes decay to-

ward zero:

$$r_1(t), r_2(t) \rightarrow 0.$$

At first this seems surprising because each isolated oscillator has a stable limit cycle when $\lambda > 0$. However, coupling changes the stability of the full coupled system. For appropriate ranges of coupling strength and frequency mismatch, the origin can become stable.

To study amplitude death, write

$$z_1 = x_1 + iy_1, \quad z_2 = x_2 + iy_2,$$

and use the real state vector

$$\mathbf{u} = (x_1, y_1, x_2, y_2)^T.$$

Linearizing the full real system at the origin gives the Jacobian

$$J_0 = \begin{pmatrix} \lambda - K & -\omega_1 & K & 0 \\ \omega_1 & \lambda - K & 0 & K \\ K & 0 & \lambda - K & -\omega_2 \\ 0 & K & \omega_2 & \lambda - K \end{pmatrix}. \quad (11)$$

Amplitude death occurs when all eigenvalues of this Jacobian have negative real parts. Equivalently, if μ_j are the eigenvalues of J_0 , the origin is linearly stable when

$$\max_j \text{Re}(\mu_j) < 0.$$

This condition also explains why a positive real eigenvalue cannot produce amplitude death. A positive real part means that perturbations grow away from the origin, so the origin is unstable. Thus, amplitude death requires the entire spectrum of the linearized coupled system to lie in the left half of the complex plane.

This also shows why the phase-only model cannot describe amplitude death. The phase reduction assumes amplitudes remain close to the limit cycle, but amplitude death is precisely the regime where the amplitudes collapse. The key mechanism is the conflict between coupling and frequency mismatch. Each oscillator tends to rotate at its own frequency, while coupling tries to force the oscillators together. In the amplitude-death regime, the stable long-time behavior is not a shared oscillation, but suppression of the oscillation itself.

G. Regime Map

The two-oscillator dynamics can be summarized using a regime map in the $(K, \Delta\omega)$ plane. The three main regimes are phase drift, phase locking, and amplitude death.

For weak coupling, the system exhibits phase drift. The amplitudes remain nonzero, but the phase difference does not settle. For stronger coupling, the system

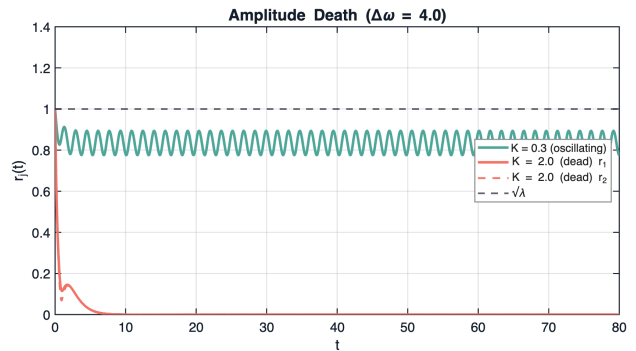


FIG. 18. Amplitude response for two coupling strengths. For weaker coupling, the oscillators continue to approach a nonzero oscillatory state near the limit-cycle radius. For stronger coupling with sufficient frequency mismatch, both amplitudes decay toward zero, indicating amplitude death. This comparison shows that amplitude death is a stability change in the full coupled system, not simply ordinary phase locking.

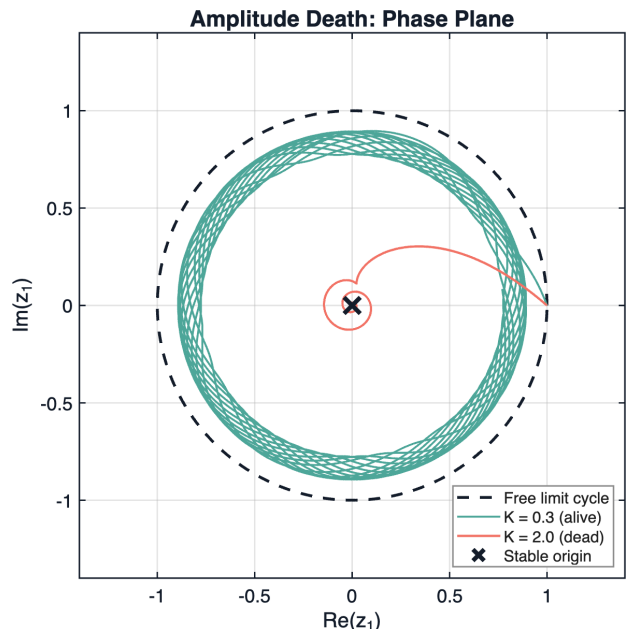


FIG. 19. Phase-plane view of amplitude death. Instead of approaching a finite-radius limit cycle, the trajectories decay toward the origin. This behavior is possible only when the full coupled-system equilibrium is stable.

can phase lock. The reduced phase model predicts the drift-locking transition near

$$K_c = \frac{|\Delta\omega|}{2}.$$

For sufficiently large mismatch and coupling values that make the origin linearly stable, the system can enter amplitude death. In this region, oscillations are suppressed and both amplitudes decay to zero. This region cannot be explained by the phase-only model because the amplitude dynamics are essential.

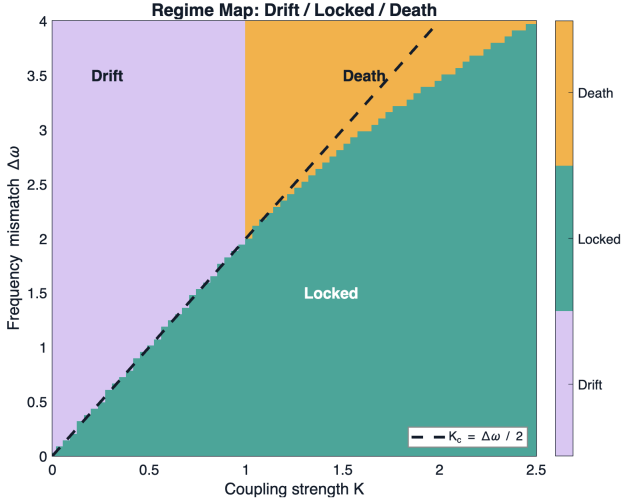


FIG. 20. Regime map for two coupled Stuart-Landau oscillators. Weak coupling produces phase drift, stronger coupling can produce phase locking, and sufficiently large frequency mismatch with stabilizing coupling can produce amplitude death. The theoretical phase-locking boundary $K_c = |\Delta\omega|/2$ comes from the reduced phase model and applies when amplitudes remain near the limit cycle.

IV. NETWORKS OF COUPLED STUART-LANDAU OSCILLATORS

The third stage is to generalize from two oscillators to a network of N coupled Stuart-Landau oscillators. We write the general network model as

$$\dot{z}_j = \left(\lambda + i\omega_j - |z_j|^2 \right) z_j + C_j(\mathbf{z}; K), \quad j = 1, \dots, N,$$

where $\mathbf{z} = (z_1, \dots, z_N)$ and $C_j(\mathbf{z}; K)$ represents the coupling term acting on oscillator j . The exact form of C_j depends on the network topology. This notation is useful because the same local Stuart-Landau dynamics are used in every case, while only the coupling structure changes.

The main question is how network topology affects synchronization. Even if every oscillator follows the same local equation, the collective behavior can change depending on which oscillators are connected and how information moves through the network.

A. Coupling Topologies

In a chain, each interior oscillator is connected to its nearest neighbors:

$$\dot{z}_j = \left(\lambda + i\omega_j - |z_j|^2 \right) z_j + K(z_{j+1} + z_{j-1} - 2z_j).$$

The endpoints have only one neighbor. A chain is a local topology, so information must travel from one node to the next. Because of this, the chain often requires a relatively large coupling strength to synchronize.

A ring is a chain with periodic boundary conditions:

$$z_{N+1} = z_1, \quad z_0 = z_N.$$

Every node has two neighbors, and there are no end-points. This makes the ring more symmetric than the chain. Ring coupling usually synchronizes more efficiently than chain coupling because information can travel in both directions around the loop.

In global coupling, every oscillator interacts with every other oscillator:

$$\dot{z}_j = \left(\lambda + i\omega_j - |z_j|^2 \right) z_j + K \sum_{k=1}^N (z_k - z_j).$$

This topology is highly connected. Each oscillator feels the influence of the entire population. As a result, global coupling tends to synchronize most efficiently. In this form, the coupling strength is not normalized by N , so the total coupling influence increases with the number of oscillators. Since N is fixed in the numerical comparisons, this choice is consistent across the topologies studied here.

In feedforward coupling, information flows mainly in one direction:

$$\begin{aligned} \dot{z}_1 &= \left(\lambda + i\omega_1 - |z_1|^2 \right) z_1, \\ \dot{z}_j &= \left(\lambda + i\omega_j - |z_j|^2 \right) z_j + K(z_{j-1} - z_j), \quad j = 2, \dots, N. \end{aligned}$$

The first oscillator acts like a leader, and each later oscillator follows the previous one. Feedforward coupling can synchronize efficiently in some cases, but it is sensitive to the ordering of the oscillators.

B. Synchronization Metrics

For two oscillators, the phase difference $\phi(t)$ is enough to study phase locking. For a network, a population-level measure is needed. The main phase-coherence metric used here is the Kuramoto-style order parameter

$$R(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right|.$$

This satisfies $0 \leq R(t) \leq 1$. If the phases are spread out around the unit circle, the complex exponentials cancel and $R(t)$ is small. If the phases are aligned, the exponentials point in nearly the same direction and $R(t)$ is close to one. This order parameter is standard in synchronization theory [2, 3].

Since $R(t)$ can fluctuate in time, especially near the transition to synchronization, we also use a time-averaged order parameter. After discarding an initial transient interval, the average coherence over the diagnostic time window $[T_1, T_2]$ is

$$\bar{R} = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} R(t) dt.$$

Coupling Topologies



FIG. 21. Comparison of coupling topologies used in the network simulations. Chain and ring coupling are local, global coupling is all-to-all, and feedforward coupling is directed. All simulations use the same local Stuart-Landau dynamics, so differences in synchronization behavior are attributed to the coupling graph.

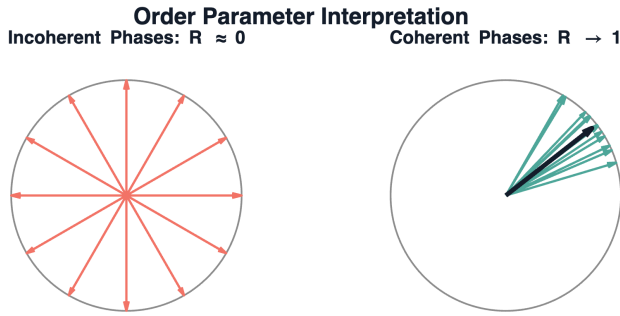


FIG. 22. Interpretation of the order parameter. When phases are spread out, the complex exponentials cancel and R is small. When phases align, the complex exponentials point in nearly the same direction and R approaches one.

In the numerical simulations, this integral is approximated by averaging the sampled values of $R(t)$ over the post-transient portion of the trajectory. Thus, $\bar{R}$ measures the sustained level of phase coherence rather than the value of $R(t)$ at one particular time.

However, phase coherence alone is not always enough. We also measure the frequency spread

$$\Delta\Omega = \max_j \Omega_j - \min_j \Omega_j,$$

where Ω_j is the long-time average frequency of oscillator j . This quantity measures whether the oscillators have reached a common average rotation rate.

In this project, network synchronization is defined using both the time-averaged order parameter and the frequency spread. A network is classified as synchronized when

$$\bar{R} > 0.90$$

and

$$\Delta\Omega < \text{tolerance}.$$

The critical coupling for a network topology is estimated as

$$K_{\text{crit}} = \min \{K : \bar{R} > 0.90 \text{ and } \Delta\Omega \approx 0\}.$$

Using both diagnostics is important. The quantity $\bar{R}$ measures how coherent the phases are on average after transients have decayed, while $\Delta\Omega$ measures whether the oscillators share a common long-time frequency. A network can have high instantaneous coherence without being frequency locked, and it can have similar average frequencies while still being spread out in phase. Requiring both large $\bar{R}$ and small $\Delta\Omega$ gives a more reliable definition of network synchronization.

C. Numerical Methods

All simulations were performed by rewriting the complex Stuart-Landau equations as real systems. For one oscillator, the real variables are defined by $z = x + iy$. For two oscillators, the state vector is

$$\mathbf{u}(t) = (x_1(t), y_1(t), x_2(t), y_2(t))^T,$$

where $z_j = x_j + iy_j$. For a network of N oscillators, the corresponding real system has dimension $2N$.

The systems were integrated over a fixed time interval where an initial transient interval was discarded before computing long-time diagnostics. For two oscillators, the main diagnostics were the amplitudes

$$r_j(t) = |z_j(t)|$$

and the phase difference

$$\phi(t) = \theta_2(t) - \theta_1(t).$$

For networks, synchronization was measured using the instantaneous order parameter $R(t)$, the time-averaged order parameter $\bar{R}$, and the long-time frequency spread $\Delta\Omega$. The value of $\bar{R}$ was computed over the post-transient time interval, so it represents sustained phase coherence rather than a single-time snapshot. A network was classified as synchronized only when it had both large time-averaged phase coherence and small frequency spread.

D. Network Results and Topology Comparison

The network simulations show that coupling topology strongly affects the coupling strength required for synchronization. The same local Stuart-Landau oscillator dynamics were used in each case, so the observed differences are caused by the coupling structure rather than by changes in the individual oscillator model.

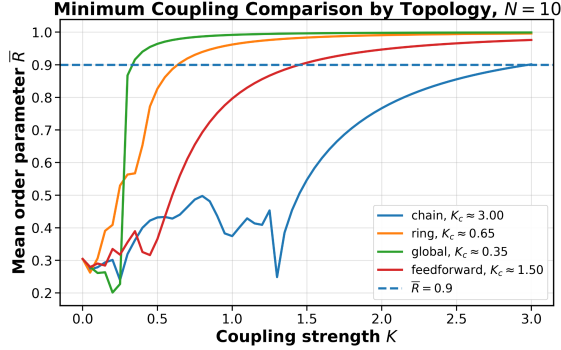


FIG. 23. Comparison of network synchronization behavior across coupling topologies. The curves show how the time-averaged order parameter $\bar{R}$ changes with coupling strength K for chain, ring, global, and feedforward coupling. The dashed horizontal line marks the coherence threshold used for classification. Topologies that cross this threshold at smaller K synchronize more efficiently, while topologies requiring larger K are less efficient at spreading phase information through the network. The $N = 10$ simulations use $\lambda = 1$ with a random initial condition, and random frequencies ω between $[0.8, 1.2]$

Figure 23 compares the time-averaged order parameter $\bar{R}$ as the coupling strength K is increased. For all topologies, increasing K generally increases phase coherence. However, the rate at which $\bar{R}$ approaches the synchronization threshold depends strongly on the topology. Global coupling reaches the coherence threshold at the smallest coupling values, ring and feedforward coupling reach the threshold at intermediate values, and chain coupling requires the largest coupling strength. Thus, the order-parameter sweep shows that more connected or more direct coupling structures organize the oscillator phases more efficiently.

Figure 24 summarizes the same trend by plotting the estimated minimum coupling strength required for synchronization. In this numerical experiment, global coupling has the smallest estimated K_{crit} , followed by ring and feedforward coupling, while chain coupling has the largest estimated K_{crit} . This agrees with the intuition that all-to-all coupling spreads phase information most directly, whereas local coupling requires information to pass through intermediate oscillators.

The chain topology is least efficient because information travels only through nearest-neighbor connections. If one end of the chain needs to influence the other end, that influence must pass through all intermediate nodes.

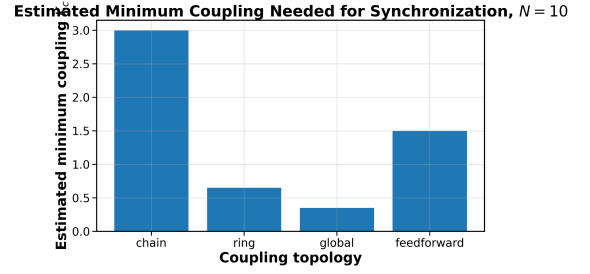


FIG. 24. Estimated minimum coupling needed for synchronization for each network topology. The bar heights summarize the approximate coupling value at which the topology satisfies the synchronization criteria. Smaller bars indicate more efficient synchronization. Global coupling requires the smallest coupling strength, while local or directed topologies require stronger coupling because phase information spreads less directly through the network. Simulations use the same $N = 10$ parameter as Figure 23.

This makes chain synchronization difficult and explains why the chain has the largest estimated coupling threshold in Fig. 24.

Ring coupling improves on the chain because there are no endpoints. Information can travel in both directions around the loop, so disturbances or phase differences can be corrected from two directions instead of one. This helps explain why the ring reaches high $\bar{R}$ at smaller K than the chain in Fig. 23.

Global coupling is most efficient in these simulations because each oscillator interacts directly with the entire population. There is no need for information to travel step by step through intermediate nodes. This direct population-level coupling causes $\bar{R}$ to rise quickly and causes the estimated K_{crit} to be the smallest among the topologies tested. Since the global coupling term is not normalized by N or by node degree, this result reflects both the all-to-all topology and the larger total coupling input experienced by each oscillator.

Feedforward coupling has a directed structure. It can synchronize efficiently when the leading oscillator is compatible with the rest of the network, but because the influence is one-way, the behavior depends on the ordering of oscillators. In the numerical results, feedforward coupling performs better than the chain but not as well as global coupling, which is consistent with the fact that feedforward coupling gives directed information transfer but not full mutual interaction.

Figure 25 confirms that phase coherence alone is not the only criterion for synchronization. The plotted frequency spread $\Delta\Omega$ decreases as coupling becomes stronger, meaning that the oscillators approach a common average rotation rate. A topology is considered synchronized only when $\bar{R}$ is large and $\Delta\Omega$ is small. Thus, Fig. 23 measures average phase alignment, while Fig. 25 checks whether the oscillators are also frequency locked.

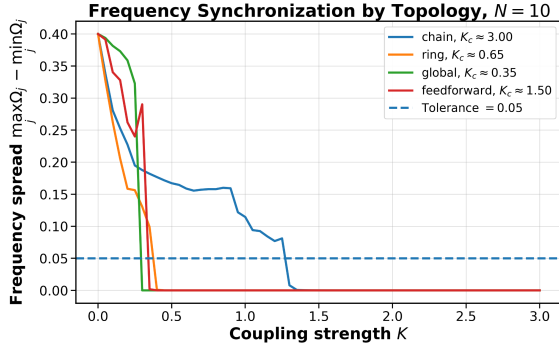


FIG. 25. Frequency-spread comparison across network topologies. The plotted quantity is $\Delta\Omega = \max_j \Omega_j - \min_j \Omega_j$, where Ω_j is the long-time average frequency of oscillator j . A synchronized network should have $\Delta\Omega$ close to zero, meaning that all oscillators share a common average rotation rate. Global coupling reduces frequency spread most rapidly, while local topologies require stronger coupling to approach frequency synchronization. Simulations use the same $N = 10$ parameter choices as Figure 23.

V. PARAMETER CHOICES AND LIMITATIONS

Each parameter in the Stuart-Landau model has a specific role. The bifurcation parameter λ determines whether an isolated oscillator has a stable limit cycle. In most simulations, $\lambda > 0$ is chosen so that each oscillator would oscillate on its own. Choosing $\lambda = 1$ ensures that each uncoupled oscillator has stable oscillatory behavior. The coupling strength of K determines how strongly oscillators influence one another and plays a major role in the transitions between drifting, locking, and synchronization. The frequencies and initial conditions also influence the long-term dynamics of the system.

The frequencies ω_j determine the natural rotation rates. If the frequencies are identical, synchronization is easier. If they are different, coupling must overcome the mismatch. For two oscillators, this mismatch is measured by $\Delta\omega = \omega_2 - \omega_1$. For networks, the spread of the natural frequencies affects how difficult synchronization is to achieve. A network with similar ω_j values is generally easier to synchronize than a network with widely separated natural frequencies.

The coupling strength K determines how strongly oscillators influence each other. Small K produces nearly independent behavior. Larger K can produce phase locking or network synchronization. For two coupled oscillators, amplitude death occurs only in parameter regimes where the origin is linearly stable. Thus, coupling and frequency mismatch must be strong enough to suppress the limit-cycle oscillations, but the stability condition is determined by the eigenvalues of the full coupled system rather than by coupling strength alone.

The topology determines how coupling is distributed across the network. The same value of K can produce different outcomes depending on whether the network is

TABLE I. Main model parameters and diagnostics used in the simulations.

Parameter	Meaning
λ	Bifurcation parameter controlling growth or decay
ω_j	Natural frequency of oscillator j
$\Delta\omega$	Natural frequency mismatch for two oscillators
K	Coupling strength
r_j	Amplitude of oscillator j
θ_j	Phase of oscillator j
ϕ	Phase difference between two oscillators
$R(t)$	Instantaneous network phase coherence
$\bar{R}$	Time-averaged network phase coherence after transients
$\Delta\Omega$	Spread of average network frequencies
K_{crit}	Estimated coupling threshold for synchronization

a chain, ring, global graph, or feedforward system. Local topologies such as chains transmit information gradually through neighboring nodes, while global coupling allows each oscillator to respond directly to the full population.

The order parameter $R(t)$ measures instantaneous phase coherence. Since $R(t)$ may fluctuate over time, especially near the synchronization transition, we use the time-averaged quantity $\bar{R}$ to classify long-time behavior. The value of $\bar{R}$ is computed only after discarding transients, so it measures the sustained coherence of the network. In this project, a network is considered phase coherent when $\bar{R} > 0.90$.

The frequency-spread diagnostic $\Delta\Omega$ measures whether oscillators share a common average frequency. This is included because large $\bar{R}$ alone does not always guarantee full synchronization. A network could be temporarily phase coherent but not frequency locked. Therefore, the synchronization criterion requires both large $\bar{R}$ and small $\Delta\Omega$.

Initial conditions also matter for numerical exploration. For the single oscillator, different initial radii show attraction toward the limit cycle. For two oscillators, different initial phases test whether the locked state is attracting. For networks, random initial phases test whether the population can organize from disorder.

There are several limitations to the analysis. First, the reduced phase model is an approximation. It assumes amplitudes remain close to the stable limit cycle. This is reasonable for phase locking, but it fails for amplitude death.

Second, the numerical synchronization criterion depends on thresholds. In this project, synchronization is defined using $\bar{R} > 0.90$ and small frequency spread. A different threshold would change the estimated critical coupling values.

Third, the choice of averaging window affects $\bar{R}$. If the transient interval is too short, the average may include early-time behavior that does not represent the final state. If the simulation time is too short, slowly converging systems near critical coupling may be misclassified.

Fourth, $\bar{R}$ is an average and can hide time-dependent

behavior. For example, two simulations could have similar values of $\bar{R}$ even if one has nearly constant coherence and the other has intermittent bursts of coherence. This is why the time series $R(t)$ and the frequency spread $\Delta\Omega$ are useful supporting diagnostics.

Fifth, the results depend on the distribution of natural frequencies. A network with small frequency spread is easier to synchronize than a network with large frequency spread.

Finally, the model is idealized. Real systems may include noise, delay, external forcing, changing topology, or nonlinear coupling. The Stuart-Landau model is valuable because it isolates the core mechanisms, but real applications may require additional effects.

VI. APPLICATIONS AND FUTURE WORK

Coupled oscillators appear in many real systems. In biology, oscillators can represent heart cells, circadian rhythms, or neural populations. Synchronization may represent coordinated beating, common rhythmic timing, or coherent brain activity. In electrical systems, oscillators can represent circuits or power-grid components. Synchronization is important because components often need to maintain common frequencies. Failure to synchronize can lead to instability or inefficient operation.

In lasers, coupled oscillators can represent interacting optical modes. Depending on the coupling and detuning, the modes may lock, drift, or suppress each other. In sensor networks, each node may have its own timing or oscillatory behavior. Synchronization can help align measurements or coordinate communication. The network topology results are especially relevant here because global coupling is efficient but may be expensive, while local coupling is easier to implement but requires stronger coupling or more time.

There are several natural extensions of this project. One extension is time-delayed coupling. In many real systems, one oscillator does not respond instantly to another. A delayed coupling term could have the form

$$z_k(t - \tau) - z_j(t),$$

where τ is the delay. Delays can create new instabilities and more complicated synchronization patterns.

Another extension is noise. A noisy Stuart-Landau oscillator could be written as

$$\dot{z} = (\lambda + i\omega - |z|^2)z + \sigma\xi(t),$$

where $\xi(t)$ represents random forcing and σ controls the noise strength. This would allow robustness of synchronization to be studied.

A third extension is a two-dimensional lattice. Instead of placing oscillators in a chain or ring, they could be placed on a grid. This could produce waves, spirals, clusters, or localized synchronization. Another direction is to allow different values of λ_j for different oscillators.

Then some oscillators might naturally oscillate while others naturally decay. Coupling could determine whether oscillation spreads through the network or dies out.

Finally, more realistic network topologies could be studied, including random graphs, small-world networks, or scale-free networks. This would connect the model more closely to biological, social, and engineered networks.

VII. SUMMARY AND CONCLUSION

The main mathematical results are summarized as follows. The single oscillator is

$$\dot{z} = (\lambda + i\omega - |z|^2)z.$$

In Cartesian coordinates,

$$\begin{aligned}\dot{x} &= (\lambda - x^2 - y^2)x - \omega y, \\ \dot{y} &= (\lambda - x^2 - y^2)y + \omega x.\end{aligned}$$

The Jacobian at the origin has eigenvalues

$$\eta_{\pm} = \lambda \pm i\omega.$$

In polar coordinates,

$$\begin{aligned}\dot{r} &= (\lambda - r^2)r, \\ \dot{\theta} &= \omega.\end{aligned}$$

For $\lambda > 0$, the stable limit-cycle radius is

$$r^* = \sqrt{\lambda}.$$

For two coupled oscillators, the reduced phase equation is

$$\dot{\phi} = \Delta\omega - 2K \sin \phi.$$

The critical coupling predicted by the reduced phase model is

$$K_c = \frac{|\Delta\omega|}{2}.$$

Amplitude death is not captured by this phase-only approximation. Instead, it is determined by the stability of the origin in the full coupled system. Using the Jacobian J_0 in Eq. (11), amplitude death requires

$$\max_j \text{Re}(\mu_j) < 0,$$

where μ_j are the eigenvalues of J_0 .

For networks, the instantaneous order parameter is

$$R(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right|.$$

TABLE II. Main regimes observed in the coupled Stuart-Landau system.

Regime	Diagnostic behavior
Stable limit cycle	$r(t) \rightarrow \sqrt{\lambda}$ for $\lambda > 0$
Phase drift	$\phi(t)$ does not approach a constant
Phase locking	$\phi(t) \rightarrow \phi^*$
Amplitude death	$r_1(t), r_2(t) \rightarrow 0$
Network synchronization	$\bar{R} > 0.90$ and small $\Delta\Omega$

The time-averaged order parameter $\bar{R}$ is computed after transients have decayed and measures sustained phase coherence. Large $\bar{R}$ and small frequency spread indicate network synchronization.

In this project, we studied the Stuart-Landau oscillator as a model for nonlinear oscillations, Hopf bifurcation, synchronization, phase locking, amplitude death, and network-level collective behavior. For the single oscillator, the origin changes stability as λ crosses zero,

and the polar equation shows that a stable limit cycle appears for $\lambda > 0$. For two coupled oscillators, the reduced phase equation shows that phase locking is possible only when the coupling is strong enough to overcome frequency mismatch. The amplitude-death analysis shows that coupling does not always synchronize oscillators; in parameter regimes where the full coupled origin becomes stable, oscillations can be suppressed entirely. Finally, the network simulations show that topology strongly affects synchronization. Global coupling synchronizes most efficiently, while chain coupling requires the largest coupling strength.

Overall, the Stuart-Landau oscillator provides a clear framework for understanding how simple nonlinear oscillators can produce rich collective behavior when coupled. The project connects Hopf bifurcation theory, phase reduction, amplitude death, and network synchronization in one compact model. The standard theory explains the single-oscillator and reduced two-oscillator behavior, while the numerical contribution compares how coupling topology changes synchronization outcomes under the same diagnostic criteria.

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